

fmix Layer 2: A Best-Effort Size Contract for Phase A

Mechanism, Controller Design, and the Running Experiments

local_mixing working notes

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Abstract

Layer 2 turns phase A from a set of hand-tuned rates into a *contract*: a caller states a size trajectory — expand to $R_1 \times$ the input by N_0 units of effective work, hold until N_1 , compress toward $R_2 \times$ within a further budget — and a controller delivers it, or says clearly why it cannot. The single lever is `p_mix`, the per-round MIX/COMP coin; the controller identifies the plant from the monitors the run already keeps, and steers. Crucially it does *not* model twists: it measures their effect as a disturbance, which is what lets a profile run at any twist rate without being told the rate. This note gives the mechanism, the design decisions (especially the anti-over-steer measures), what validation established, and the eight-arm experiment now running on a 100k-gate, 512-wire slice. Headline from the runs so far: the four R_1 arms track their setpoints inside $\pm 3\%$ across expansion and hold, and there is a hard *twist ceiling* — at `p_twist` = 5×10^{-3} no size contract is achievable at any lever setting.

1 What layer 2 is

1.1 The phase-A defaults (-phase-a)

A single flag fixes the phase-A operating point: g57-word twist brackets (`-twist-g57`) at `p_twist` = 5×10^{-4} ; `-db-advance` on (per the standing directive — DB splice products must ride their assigned direction); equal parts convex and contiguous window sampling (`p_convex` = 0.5); MIX with `p_mingen` = 0.6 and the curated cascade (`-curated`, `-mix-pay-random`); COMP with `p_mingen` = 0 and the regular store only. Any flag given explicitly on the command line wins, and the startup banner prints the *resolved* values, so a run states what it actually is.

`-mix-pay-random` is the one selection change: when a MIX window's store answer contains only *larger* spellings, take a uniformly random one rather than a minimal one. This buys growth and spelling diversity per paid splice — and gives the controller a stronger up-lever.

1.2 The size profile (-profile N0,N1,N2,R1,R2)

Three phases, in effective-work units E (cumulative moves per gate — the same clock all our analysis uses, so the contract is size-independent):

1. **expand** to $R_1 \cdot S_{\text{in}}$ by $E = N_0$;
2. **hold** near $R_1 \cdot S_{\text{in}}$ until $E = N_1$;
3. **compress** toward $R_2 \cdot S_{\text{in}}$, ending on arrival or at N_2 — whichever comes first.

The setpoint $S^*(E)$ is the corresponding piecewise-linear trajectory. N_2 may be an absolute mark ($N_2 \geq N_1$) or the compression leg’s *budget*; a value below N_1 can only mean the latter, and the run logs the absolute end mark it derived. A completed schedule *ends the run* (`MixStop::ProfileDone`) rather than burning the remaining move budget outside any setpoint — which, at a twist rate the lever cannot offset, actively undoes the compression leg (measured: +309% over target before this stop existed).

1.3 One size authority

`fmix` previously had three mechanisms pushing size around — the thermostat’s static `target_size`, the size brake’s `size_hi/size_lo` guard rails, and the `p_mix` coin. Adding a controller as a fourth would have produced two controllers on one plant: the profile pressing the accelerator through phase 1 while the brake (seeing “runaway growth”) and the thermostat (still targeting the input size) pressed back — and the plant estimates would have been polluted by their interference. So while a profile is active it is the *only* size authority: it owns `target_size`, continuously setting it to $S^*(E)$ so the thermostat is conscripted rather than disabled, and the static brake is inert. Passing `-target-size`, `-size-hi`, `-size-lo` or `-p-mix` alongside `-profile` is refused outright rather than silently overridden.

2 The controller

2.1 Plant identification

Every `-prof-cadence-eff` of effective work (default 0.5) the controller reads the run’s own counters and estimates three quantities, all in *gates per move*:

- $\hat{g}$ — drift if the lever were pinned at 1 (all MIX);
- $\hat{s}$ — removal rate if the lever were pinned at 0 (all COMP);
- $\hat{d}$ — the **disturbance**: the residual between the observed total drift and the drift the DB move accounts for.

$\hat{g}$ and $\hat{s}$ come from per-mode gate-delta counters added for this purpose, normalised by the lever actually in force over the interval (an arm that saw too little of the interval keeps its previous estimate rather than dividing by ≈ 0). All three are EWMA-smoothed (`-prof-ewma`).

2.2 Why the disturbance term is the whole trick

The controller cannot steer the twist rate — twists are a layer-1 mechanism with their own purpose — so it must *cope* with it. Rather than modelling twist growth (which would need the rate, the bracket cost, the absorption statistics...), the controller measures what it does: everything that changes size and is not the DB move lands in $\hat{d}$. Twists dominate that residual; expansion moves and thermostat contractions also live there. The lever then solves

$$p\hat{g} - (1-p)\hat{s} + \hat{d} = v^*,$$

where v^* is the slope that lands on the setpoint one cadence ahead, plus a small integral term on the tracking error. A twist-heavy run simply gets a lower `p_mix`, automatically, with no knowledge of the twist rate.

The running experiments validate the mechanism directly: $\hat{d} = +0.0000$ with twists off, $+0.0042$ at `p_twist` = 5×10^{-4} , $+0.0388$ at 5×10^{-3} .

2.3 Not over-sampling, not over-steering

Measurement is $O(1)$ counter reads — there is no census cost — so “over-sampling” is purely a statistical concern, addressed by the cadence and the EWMA. Over-steering is the real risk, and four guards address it:

- **cadence in effective-work units**, so control frequency scales with circuit size and each interval contains tens of thousands of rounds;
- **deadband** (`-prof-deadband`, default 2%): no lever change while the size is close to setpoint;
- **rate limit** (`-prof-dp-max`, default 0.1 per update) — with an *escape hatch*: the limit is lifted while more than four deadbands from the setpoint, because a 0.1-per-step crawl cannot catch a ramp only six effective-work units long. Gentle where gentleness matters, free where it does not;
- **phase hysteresis**: the integral term and saturation counter reset at each phase boundary.

2.4 Best effort, stated honestly

Not every profile is reachable. When the lever pins and the size is still clearly off setpoint, the controller logs `profile: SATURATED` and holds the pinned lever — it does not thrash, and it does not abort. Phase 3 additionally diagnoses the specific infeasibility that matters: when $\hat{d} > \hat{s}$, the disturbance exceeds the maximum available removal, so the circuit grows with the lever at 0 no matter what, and the log says exactly that (*lower the twist rate or relax R_2*). Every control point prints `phase`, `eff`, `size`, S^* , `pmix`, $\hat{g}$, $\hat{s}$, $\hat{d}$, the integrator and the saturation counter, so a run’s adherence to its contract is auditable after the fact.

3 What validation established

3.1 The plant

On the 100k gadget at the phase-A operating point: $\hat{g} \approx 1.7\text{--}1.8$ gates/move at full lever — consistent with the parallel session’s independent `p_mix` sweep (final size ratio $1.28\times \rightarrow 3.27\times$ over `p_mix` $0.05 \rightarrow 0.40$, monotone and smooth). Critically, $\hat{s}$ *decays* as material hardens: from ≈ 0.5 early to ≈ 0.005 late in a 20k run. A static feed-forward table calibrated on early drift would therefore overshoot badly late in a run — live estimation is not a refinement here, it is a requirement.

3.2 Two defects the first runs exposed

Early overshoot. A `p_mix= 1.0` prior blew past the expansion ramp before the first update (measured 185k against a 118k setpoint). Fixed by a moderate 0.5 prior, a first update at quarter cadence, and the escape hatch above.

Idling past completion. Without the profile-complete stop, a finished schedule kept mixing outside any setpoint; at a twist rate the lever could not offset, it grew to +309% over target.

3.3 Two preset defects found by reading a live banner

`-phase-a` decided “did the user specify this?” by comparing each knob to its default *value*, so `-p-twist 0` — a real request, and also the default — was silently overwritten with 5×10^{-4} : an intended no-twist arm ran as an exact duplicate of its baseline. It now keys on whether the flag appeared on the command line. Separately, `p_mingen` ships at 0.8, not the 0.6 phase A calls for, and a stale comment led the preset to skip setting it. Both fixed; the banner now prints resolved values.

4 The running experiments

Eight arms, all from the same 100k-gate, 512-wire slice, sampled ancestry ($K = 256$), measurements every 200k moves. Four are the requested R_1 sweep; four isolate one axis each against the `nR20` baseline.

arm	profile	p_twist	what it isolates
nR15	5, 50, +20, 1.5, 1.2	5×10^{-4}	expansion ratio
nR20	5, 50, +20, 2.0, 1.2	5×10^{-4}	baseline
nR25	5, 50, +20, 2.5, 1.2	5×10^{-4}	expansion ratio
nR30	5, 50, +20, 3.0, 1.2	5×10^{-4}	expansion ratio
notwist	5, 50, +20, 2.0, 1.2	0	twist axis
nR20_tw5	5, 50, +20, 2.0, 1.2	5×10^{-3}	twist axis
nR20_fast	5, 20, +20, 2.0, 1.2	5×10^{-4}	hold duration
nR20_slow	5, 100, +20, 2.0, 1.2	5×10^{-4}	hold duration

Table 1: The question is which shape buys the most mixing quality per unit of work — read from the sampled-ancestry measures (`desc`, `cov`, `ent`) along the way and from affine-reconstruction plates on the finals.

4.1 Tracking, in progress

arm	phase	E	size	S^*	error	p_mix
nR15	hold	42.2	147,766	150,000	−1.5%	0.000
nR20	hold	28.2	193,975	200,000	−3.0%	0.091
nR25	hold	22.2	245,538	250,000	−1.8%	0.111
nR30	hold	18.3	295,169	300,000	−1.6%	0.085
nR20_slow	hold	28.2	193,975	200,000	−3.0%	0.091
nR20_fast	compress	28.9	168,584	164,473	+2.5%	0.000
nR20_tw5	hold	39.1	368,552	200,000	+84%	0.000

Six of seven track inside $\pm 3\%$, including `nR20_fast` descending its compression ramp. (`nR20` and `nR20_slow` agree exactly: same seed, and their setpoints coincide until $E = 50$ — a useful determinism check.)

4.2 The twist ceiling

`nR20_tw5` is the informative failure. At $p_twist = 5 \times 10^{-3}$ the controller cannot even *hold*: the lever is pinned at 0 and the circuit still runs 84% above setpoint, because $\hat{d} = +0.039$ gates/move exceeds the maximum available removal $\hat{s} = +0.021$. This reproduces a 20k-scale result exactly,

and it is a genuine constraint rather than a controller deficiency: **a size contract and a high twist rate are mutually exclusive**. Choose the twist rate first, then ask for a profile the plant can deliver.

Layer-2 profile controller on the 100k / 512-wire slice (runs in progress)

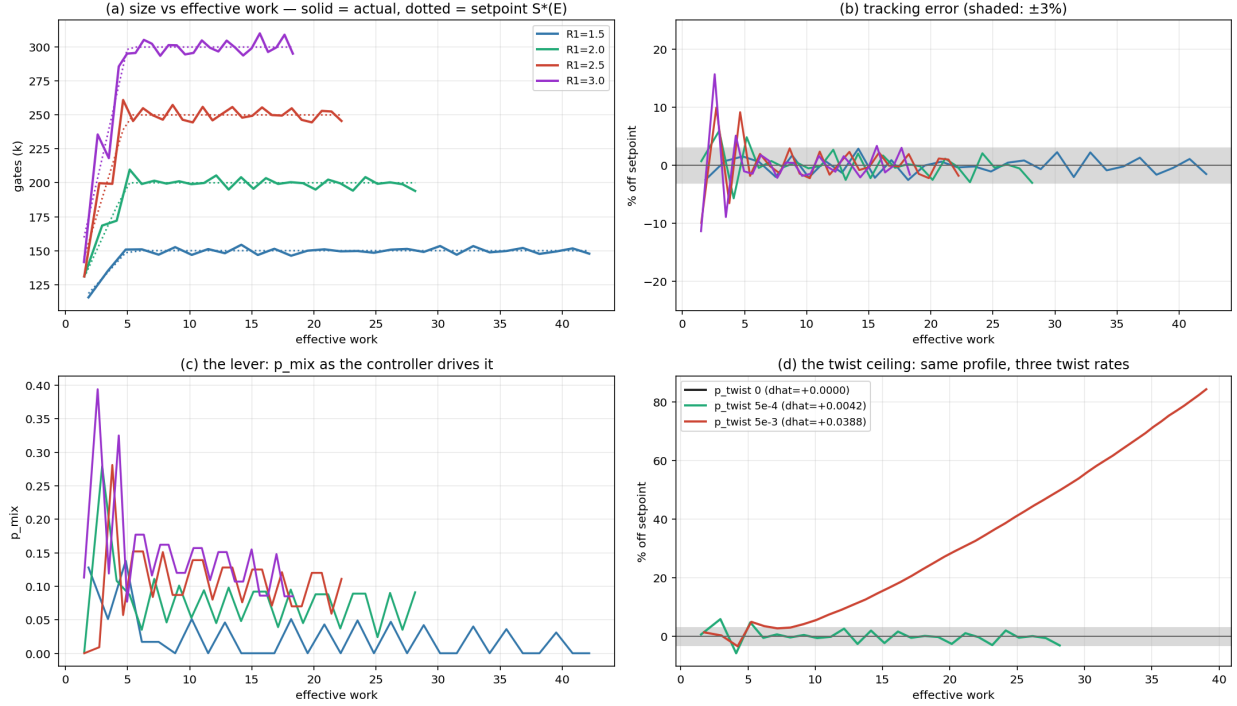


Figure 1: (a) size against effective work, actual (solid) versus setpoint (dotted), for the four R_1 arms; (b) tracking error, shaded band $\pm 3\%$; (c) the lever the controller chose; (d) the twist axis — same profile at three twist rates, with the measured disturbance in the legend.

5 Caveats and open items

- The running arms use `p_mingen = 0.8`, not the phase-A 0.6, because the preset defect above was found after they launched. Every arm shares it, so the *comparison between profiles* is sound; the absolute operating point is not exactly phase A. The intended remedy is to rerun only the winning profile at the exact spec.
- A profile is a whole-run construct: its effective-work clock is not serialised, so `-profile` on a `-resume` restarts the schedule at $E = 0$ (warned at startup).
- Single trajectories, one seed per arm — these are engineering measurements, not replicated statistics.
- The mixing-quality comparison (which profile shape is best) is the point of the campaign and is still pending: the arms are mid-flight.